

Linearization of the prediction equation

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}, \quad x(t) \in \mathbb{R}^n. \quad (1)$$

$$\chi(t) = \begin{bmatrix} x(t) \\ 1 \end{bmatrix}. \quad (2)$$

$$\begin{aligned} \dot{\chi}(t) &= \begin{bmatrix} \dot{x}(t) \\ \frac{d}{dt}(1) \end{bmatrix} \\ &= \begin{bmatrix} A_{\sigma(t)}x(t) + b_{\sigma(t)} \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} A_{\sigma(t)} & b_{\sigma(t)} \\ 0_{1 \times n} & 0 \end{bmatrix} \begin{bmatrix} x(t) \\ 1 \end{bmatrix}. \end{aligned} \quad (3)$$

$$\tilde{A}_{\sigma(t)} = \begin{bmatrix} A_{\sigma(t)} & b_{\sigma(t)} \\ 0_{1 \times n} & 0 \end{bmatrix}. \quad (4)$$

$$\boxed{\dot{\chi}(t) = \tilde{A}_{\sigma(t)}\chi(t)}. \quad (5)$$

$$0 = t_0 < t_1 < t_2 < \dots < t_N, \quad (6)$$

$$\sigma(t) = q_i, \quad t_{i-1} \leq t < t_i, \quad i = 1, 2, \dots, N. \quad (7)$$

$$F_i = \tilde{A}_{q_i}, \quad i = 1, 2, \dots, N. \quad (8)$$

$$\Delta t_i = t_i - t_{i-1}. \quad (9)$$

$$\chi(t_i) = e^{F_i \Delta t_i} \chi(t_{i-1}). \quad (10)$$

$$\boxed{\chi(t_N) = e^{F_N \Delta t_N} e^{F_{N-1} \Delta t_{N-1}} \dots e^{F_2 \Delta t_2} e^{F_1 \Delta t_1} \chi(t_0)}. \quad (11)$$

$$\Delta \bar{t}_i = \bar{t}_i - \bar{t}_{i-1}, \quad i = 1, 2, \dots, N. \quad (12)$$

$$\Delta t_i = \Delta \bar{t}_i + \delta t_i. \quad (13)$$

$$\boxed{\bar{\chi}(t_N) = e^{F_N \Delta \bar{t}_N} e^{F_{N-1} \Delta \bar{t}_{N-1}} \dots e^{F_2 \Delta \bar{t}_2} e^{F_1 \Delta \bar{t}_1} \bar{\chi}(t_0)}. \quad (14)$$

$$e(t) = \chi(t) - \bar{\chi}(t). \quad (15)$$

$$\chi(t_0) = \bar{\chi}(t_0) + e(t_0). \quad (16)$$

$$\begin{aligned} e(t_N) &= e^{F_N(\Delta \bar{t}_N + \delta t_N)} e^{F_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})} \dots \\ &\quad e^{F_2(\Delta \bar{t}_2 + \delta t_2)} e^{F_1(\Delta \bar{t}_1 + \delta t_1)} (\bar{\chi}(t_0) + e(t_0)) \\ &\quad - e^{F_N \Delta \bar{t}_N} e^{F_{N-1} \Delta \bar{t}_{N-1}} \dots e^{F_2 \Delta \bar{t}_2} e^{F_1 \Delta \bar{t}_1} \bar{\chi}(t_0). \end{aligned} \quad (17)$$

$$\frac{d}{ds} e^{F_i s} = F_i e^{F_i s}. \quad (18)$$

$$e^{F_i(\Delta \bar{t}_i + \delta t_i)} = e^{F_i \Delta \bar{t}_i} + F_i e^{F_i \Delta \bar{t}_i} \delta t_i + \mathcal{O}(\delta t_i^2). \quad (19)$$

$$\phi_i = e^{F_i \Delta \bar{t}_i}. \quad (20)$$

$$e^{F_i(\Delta \bar{t}_i + \delta t_i)} = \phi_i + F_i \phi_i \delta t_i + \mathcal{O}(\delta t_i^2). \quad (21)$$

$$\begin{aligned} &e^{F_N(\Delta \bar{t}_N + \delta t_N)} e^{F_{N-1}(\Delta \bar{t}_{N-1} + \delta t_{N-1})} \dots e^{F_1(\Delta \bar{t}_1 + \delta t_1)} \\ &\doteq (\phi_N + F_N \phi_N \delta t_N) (\phi_{N-1} + F_{N-1} \phi_{N-1} \delta t_{N-1}) \dots (\phi_1 + F_1 \phi_1 \delta t_1). \end{aligned} \quad (22)$$

$$\begin{aligned}
& (\phi_N + F_N \phi_N \delta t_N) (\phi_{N-1} + F_{N-1} \phi_{N-1} \delta t_{N-1}) \cdots (\phi_1 + F_1 \phi_1 \delta t_1) \\
& \doteq \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 \\
& \quad + F_N \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 \delta t_N \\
& \quad + \phi_N F_{N-1} \phi_{N-1} \cdots \phi_2 \phi_1 \delta t_{N-1} \\
& \quad + \phi_N \phi_{N-1} F_{N-2} \phi_{N-2} \cdots \phi_2 \phi_1 \delta t_{N-2} \\
& \quad + \cdots \\
& \quad + \phi_N \phi_{N-1} \cdots \phi_3 F_2 \phi_2 \phi_1 \delta t_2 \\
& \quad + \phi_N \phi_{N-1} \cdots \phi_3 \phi_2 F_1 \phi_1 \delta t_1.
\end{aligned} \tag{23}$$

$$\begin{aligned}
e(t_N) & \doteq \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 (\bar{\chi}(t_0) + e(t_0)) \\
& \quad + F_N \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 (\bar{\chi}(t_0) + e(t_0)) \delta t_N \\
& \quad + \phi_N F_{N-1} \phi_{N-1} \cdots \phi_2 \phi_1 (\bar{\chi}(t_0) + e(t_0)) \delta t_{N-1} \\
& \quad + \cdots \\
& \quad + \phi_N \phi_{N-1} \cdots \phi_3 \phi_2 F_1 \phi_1 (\bar{\chi}(t_0) + e(t_0)) \delta t_1 \\
& \quad - \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 \bar{\chi}(t_0).
\end{aligned} \tag{24}$$

$$\begin{aligned}
e(t_N) & \doteq \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 e(t_0) \\
& \quad + F_N \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 \bar{\chi}(t_0) \delta t_N \\
& \quad + \phi_N F_{N-1} \phi_{N-1} \cdots \phi_2 \phi_1 \bar{\chi}(t_0) \delta t_{N-1} \\
& \quad + \phi_N \phi_{N-1} F_{N-2} \phi_{N-2} \cdots \phi_2 \phi_1 \bar{\chi}(t_0) \delta t_{N-2} \\
& \quad + \cdots \\
& \quad + \phi_N \phi_{N-1} \cdots \phi_3 F_2 \phi_2 \phi_1 \bar{\chi}(t_0) \delta t_2 \\
& \quad + \phi_N \phi_{N-1} \cdots \phi_3 \phi_2 F_1 \phi_1 \bar{\chi}(t_0) \delta t_1.
\end{aligned} \tag{25}$$

$$\Phi = \phi_N \phi_{N-1} \cdots \phi_2 \phi_1. \tag{26}$$

$$\delta \mathbf{t} = \begin{bmatrix} \delta t_1 \\ \delta t_2 \\ \vdots \\ \delta t_{N-1} \\ \delta t_N \end{bmatrix}. \tag{27}$$

$$\begin{aligned}
\Gamma = & \begin{bmatrix} \phi_N \phi_{N-1} \cdots \phi_3 \phi_2 F_1 \phi_1 \bar{\chi}(t_0), \\ \phi_N \phi_{N-1} \cdots \phi_3 F_2 \phi_2 \phi_1 \bar{\chi}(t_0), \\ \phi_N \phi_{N-1} \cdots \phi_4 F_3 \phi_3 \phi_2 \phi_1 \bar{\chi}(t_0), \\ \cdots, \\ \phi_N F_{N-1} \phi_{N-1} \cdots \phi_2 \phi_1 \bar{\chi}(t_0), \\ F_N \phi_N \phi_{N-1} \cdots \phi_2 \phi_1 \bar{\chi}(t_0) \end{bmatrix}.
\end{aligned} \tag{28}$$

$$\boxed{e(t_N) \doteq \Phi e(t_0) + \Gamma \delta \mathbf{t}}. \tag{29}$$

Notes on corrections

1. The interval matrix is defined as $F_i = \tilde{A}_{q_i}$, where q_i is active on $[t_{i-1}, t_i)$. Writing $F_i = \tilde{A}_{\sigma(t_i)}$ is ambiguous for a right-continuous switching signal because $\sigma(t_i)$ normally belongs to the following interval.
2. The derivative in (19) uses the augmented matrix F_i . The physical matrix $A_{\sigma(t)}$ has the wrong dimension for the augmented exponential.
3. The matrix order in (23), (25), and (28) has been corrected. For example, the coefficient of δt_{N-1} is

$$\phi_N F_{N-1} \phi_{N-1} \cdots \phi_1 \bar{\chi}(t_0),$$

not

$$F_{N-1} \phi_N \phi_{N-1} \cdots \phi_1 \bar{\chi}(t_0).$$

Matrices belonging to different modes do not generally commute.

4. The columns of Γ are written horizontally and in the same order as the entries of $\delta \mathbf{t}$. This removes the dimensional ambiguity in the transposed stacked-vector notation.
5. The symbol $\doteq$ means equality up to first order around

$$e(t_0) = 0, \quad \delta \mathbf{t} = 0.$$

Products such as $\delta t_i \delta t_j$ and $\delta t_i e(t_0)$ are omitted.

6. Each δt_i in this document is a change in an interval duration. It is not yet an independent switching-instant offset. If the initial and final cycle boundaries are fixed, the duration changes must also satisfy

$$\delta t_1 + \delta t_2 + \cdots + \delta t_N = 0.$$